

AMENDMENT NO. 5 SEPTEMBER 2022

TO

IS 3589 : 2001 STEEL PIPES FOR WATER AND SEWAGE (168.3 TO 2 540 mm OUTSIDE DIAMETER) – SPECIFICATION (Third Revision)

(Amendment No. 3, line 1 to 4, Title and General) – Delete line 1 to line 4.

(Page 3, Table 2, read with Amendment No. 1 and Amendment No. 4) — Add the following notes in the table:

- 1 Elements not quoted on this table are not to be intentionally added other than for the purpose of finishing the heat.
- 2 For semi-killed quality silicon content shall be 0.08 percent maximum.
- 3 When the steel is aluminium-killed, the total aluminium content shall not be less than 0.02 percent. When the steel is silicon killed, the silicon content shall not be less than 0.10 percent. When the steel is silicon-aluminium- killed, the silicon content shall not be less than 0.03 percent and the total aluminium content shall not be less than 0.01 percent.
- 4 Micro alloying may be subject to mutual agreement between the purchaser and the supplier. Micro alloying elements like Nb, V or Ti when used individually or in combination the total content shall not exceed 0.20 percent.
- 5 Nitrogen content of the steel shall not exceed 0.012 percent which shall be ensured by the manufacturer by occasional check analysis.
- 6 For permissible deviations on product analysis, *see* Table 3.
- 7 For each reduction by 0.01 percent Carbon below the specified maximum, an increase of 0.05 percent above the maximum of manganese is permissible up to a maximum of 1.50 for grades Fe 410 and 450.'

(Page 4, clause 10.2) — Substitute the following for the existing clause:

'10.2 The hydraulic test may be substituted by a non-destructive test if agreed between manufacturer and purchaser.'

(Page 4, clause 11.1, Note) — Substitute the following note for the existing note

'NOTE — Table 5 gives preferred outside diameters and thicknesses of pipes. Other sizes of pipes not included in the table may be supplied as per the agreement between the manufacturer and the supplier, however, the agreed diameter shall be between 168.3 mm and 2540 mm and the agreed thickness shall not be less than the minimum thickness specified against each diameter mentioned in the table. In case agreed diameter of supplied pipe is other than mentioned in Table 5, the thickness of the supplied pipe shall not be less than the minimum thickness of pipe of next higher diameter mentioned in Table 5. For all such cases, the mass of steel pipes shall be calculated by the formula given in 11.1.1.'